Introduction to Superconducting Qubit Engineering

Deval Deliwala

October 16, 2024

In superconducting qubit processors, information is stored in the quantum degrees of freedom of nanofabricated, *anharmonic* oscillators constructed from superconducting circuit elements. The energy-level spectra of s.c. qubits are governed by circuit element parameters and thus are configurable. They have a rich parameter space of possible qubit properties and operation regimes, with a predictable performance, tunable to exhibit "atom-like" properties – an *artificial atom*.

1 Engineering Quantum Circuits

In this section, we will demonstrate how s.c. quantum systems can be engineered to achieve certain desired properties.

1.1 Quantum Harmonic Oscillator → Transmon Qubit

A quantum system is governed by the time-dependent Schrödinger equation,

$$\hat{H}|\psi(t)\rangle = i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle,$$

where $|\psi(t)\rangle$ is the state of the quantum system at time t and $\hat{H}$ is the Hamiltonian that describes the total energy of the system. Solving the differential equation yields

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle.$$

The time-independent Hamiltonian $\hat{H}$ is the generator for time-evolution via the operator $e^{-i\hat{H}t/\hbar}$. Determining the Hamiltonian of a system is therefore the first step to deriving its dynamical behavior. To understand the dynamics of a s.c. qubit circuit, we start with the classical description of a linear LC resonant circuit.

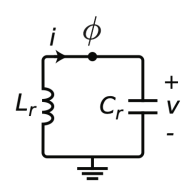

Figure 1: Simple LC resonant circuit.

Here, the energy oscillates between electrical energy in the capacitor C and magnetic energy in the inductor L. We will arbitrarily associate the electrical energy with the "kinetic energy" and the magnetic energy with the "potential energy" of the oscillator. The instantaneous, time-dependent energy in each element is derived from its current and voltage,

$$E(t) = \int_{-\infty}^{t} V(t')I(t')\,dt', \tag{1}$$

where $V(t')$ and $I(t')$ denote the voltage and current of the capacitor or inductor. To derive the classical Hamiltonian, we use the Lagrange-Hamilton formulation. Here, we represent the circuit elements in terms of one of its generalized circuit coordinates – charge or flux. We choose flux, defined as the time integral of the voltage

$$\Phi(t) = \int_{-\infty}^{t} V(t')\, dt'.$$

Here, the voltage at the node is also the branch voltage across the element. Combining the above equations for $\Phi(t)$ and $E(t)$, and using the relations

$$V = L\frac{dI}{dt} \qquad I = C\frac{dV}{dt},$$

we write down energy terms for the capacitor and inductor in terms of the node flux,

$$\tau_C = \frac{1}{2}C\dot{\Phi}^2, \tag{2}$$

$$U_L = \frac{1}{2L}\Phi^2. \tag{3}$$

The Lagrangian is defined as the difference between the kinetic and potential energy terms and can thus be expressed as

$$\mathcal{L} = \tau_C - U_L = \frac{1}{2}C\dot{\Phi}^2 - \frac{1}{2L}\Phi^2.$$

Then, using the Legendre transformation, for which we need to calculate the momentum conjugate to the flux, which in this case is the charge on the capacitor,

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi}$$

The Hamiltonian of the system is now defined as

$$H = Q\dot{\Phi} - \mathcal{L} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \equiv \frac{1}{2}CV^2 + \frac{1}{2}LI^2, \tag{4}$$

as one would expect for an electrical LC circuit. This Hamiltonian is analogous to that of a mechanical harmonic oscillator with mass $m = C$ and a resonant frequency $\omega = 1/\sqrt{LC}$, which expressed in position x, and momentum, p coordinates takes the form $H = \frac{p^2}{2m} + m\omega^2 x^2/2$.

However, the Hamiltonian described above is classical. To proceed to a quantum-mechanical description, we promote the charge and flux coordinates to quantum operators. And whereas the classical coordinates satisfy the Poisson bracket:

$$\{f, g\} = \frac{\delta f}{\delta \Phi}\frac{\delta g}{\delta Q} - \frac{\delta g}{\delta \Phi}\frac{\delta f}{\delta Q}$$
$$\rightarrow \{\Phi, Q\} = \frac{\delta \Phi}{\delta \Phi}\frac{\delta Q}{\delta Q} - \frac{\delta Q}{\delta \Phi}\frac{\delta \Phi}{\delta Q} = 1 - 0 = 1,$$

the quantum operators similarly satisfy a commutation relation:

$$[\hat{\Phi}, \hat{Q}] = \hat{\Phi}\hat{Q} - \hat{Q}\hat{\Phi} = i\hbar \tag{5}$$

Notably the same as the commutation of position and momentum, $[\hat{x}, \hat{p}]$. We can take $\hat{\Phi} \rightarrow \Phi$ and $\hat{Q} \rightarrow i\hbar\frac{d}{d\Phi}$ to yield the quantum operators $\hat{Q}$ and $\hat{\Phi}$ analogous to position and momentum.

In a simple LC resonant circuit, shown in Figure 1, both the inductor L and the capacitor C are linear circuit elements. Defining the reduced flux $\phi \equiv 2\pi\Phi/\Phi_0$ and the reduced charge $n = Q/2e$, we can write down the following quantum-mechanical Hamiltonian for the circuit,

$$H = 4E_C n^2 + \frac{1}{2}E_L\phi^2 \tag{6}$$

where $E_C = e^2/(2C)$ is the charging energy required to add *each* electron of the Cooper-pair to the island and $E_L = (\Phi_0/2\pi)^2/L$ is the inductive energy, where $\Phi_0 = h/(2e)$ is the superconducting magnetic flux quantum. The quantum operator n is the excess number of Cooper-pairs on the island, and ϕ – the reduced flux – is denoted the "gauge-invariant phase" across the inductor. These two operators obey the commutation relation $[\phi, n] = i$. The factor 4 in front of the charging energy E_c is purely a historical artifact, first defined for single-electron systems and afterwards adopted to two-electron Cooper-pair systems.

The Hamiltonian in Equation 6. is identical to the quantum harmonic oscillator. We can treat ϕ as the generalized position coordinate making the first term the kinetic energy and the second term potential energy. The functional form of the potential energy influences the possible energy eigenstates. The fact that the potential energy term is quadratic ($U_L \propto \phi^2$) in Equation 6. gives rise to the shape of the potential below.

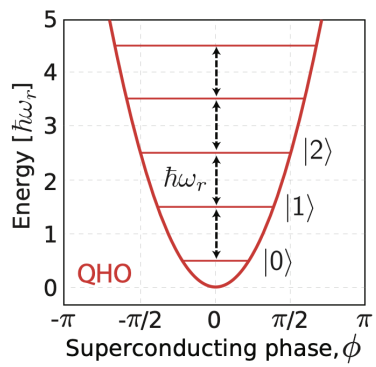

Figure 2: Energy potential for the quantum harmonic oscillator, whose energy eigenstates are equidistantly spaced $\hbar\omega_r$ apart.

The solution to the eigenvalue problem gives an infinite series of possible eigenstates $|k\rangle, (k = 0, 1, 2, \ldots)$, whose corresponding eigenenergies E_k are all equidistantly spaced, i.e. $E_{k+1} - E_k = \hbar\omega_r$, where $\omega_r = \sqrt{8E_L E_C}/\hbar = 1/\sqrt{LC}$ denotes the resonant frequency of the system. We can represent these results in a more compact form for the quantum harmonic oscillator (QHO) Hamiltonian using creation and annihilation operators:

$$H = \hbar\omega_r \left(a^\dagger a + \frac{1}{2} \right) \tag{7}$$

where $a^\dagger a$ is the creation (annihilation) operator of a single excitation of the resonator. The Hamiltonian in Equation 7. is written as an energy. From here on, $\hbar$ will be omitted; using natural units.

The linear characteristics of the QHO has a liimitation in its applications for quantum information processing. We need to be able to define a computational subspace consisting of only two energy states (usually the two lowest energy eigenstates) in between which transitions can be driven without also exciting other levels in the system. Because many gate operations depend on frequency selectivity, the equidistant level-spacing of the QHO is not possible to work with.

Therefore, we add anharmonicity (nonlinearity) into our system. We require the transition frequencies $\omega_q^{0\to1}$ and $\omega_q^{1\to2}$ to be different enough to be individually addressable. In general, the larger the anharmonicity the better, however, the amount affects how short pulses used to drive the qubit can be.

To introduce anharmonicity, we replace the linear inductor of the QHO with a Josephson junction – a nonlinear, dissipationless circuit element that forms the backbone of superconducting circuits. The potential energy of the Josephson junction can be derived similarly using Equation 1. The two Josephson junction relations

$$I = I_c \sin(\phi), \quad V = \frac{\hbar}{2e}\frac{d\phi}{dt},$$

result in the modified Hamiltonian

$$H = 4E_C n^2 - E_J \cos(\phi) \tag{8}$$

where $E_C = e^2/(2C_\Sigma), C_\Sigma = C_s + C_J$ is the total capacitance, including both shunt capacitance C_s and the self-capacitance of the junction C_J, and $E_J = I_C\Phi_0/2\pi$ is the Josephson energy, with I_C being the critical current of the junction.

The potential energy no longer takes a pure parabolic form, but rather features a cosinusoidal form from the second term in Equation 8., which makes the energy spectrum nondegenerate. This allows us to make the oscillator anharmonic and thus allows us to identify a uniquely addressable two-level quantum system shown below.

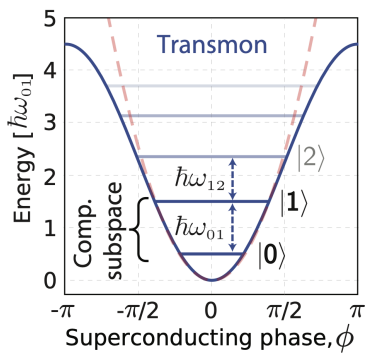

Figure 3: The Josephson junction reshapes the quadratic energy potential (dashed red) into sinusoidal (solid blue), which yields non-equidistant energy levels. This allows us to isolate the two lowest energy levels $|0\rangle$ and $|1\rangle$, forming a computational basis with energy separation $\hbar\omega_{01}$, which is different than $\hbar\omega_{12}$.

The system dynamics are governed by the dominant energy in Equation 8., reflected in the energy E_J/E_C ratio. The s.c. qubit community has converged towards circuitry with $E_J \gg E_C$, as $E_J \leq E_C$ results in qubits becoming highly sensitive to charge noise which is more difficult to deal with than flux noise. Additionally there is more flexibility in engineering the inductive (potential) part of the Hamiltonian.

To make $E_J \gg E_C$, one proffered approach is to make E_C small by shunting the junction with a large capacitor, $C_s \gg C_J$, effectively making the qubit less sensitive to charge noise – a circuit commonly known as the *transmon* qubit. Here, the superconducting phase ϕ can be used as a quantum number because the spread (or *quantum fluctuation*) of ϕ values represented by the wave function is small. The low-energy eigenstates are therefore, to a good approximation, localized states in the potential well.

If excitation to higher non-computational states is suppressed over any gate operations, either due to a large enough $|\alpha| = |\omega_q^{1\to2} - \omega_q^{0\to1}|$ or due to any robust control techniques, we can effectively treat the anharmonic oscillator (AHO) as a quantum two-level system, simplifying the Hamiltonian to

$$H = \omega_q \frac{\sigma_z}{2} = \omega_q \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \tag{9}$$

where σ_z is the Pauli-z operator. However, higher levels are often useful to implement more efficient gate operations and should not be ignored.

1.2 Qubit Hamiltonian Engineering

1.2.1 Tunable qubit: split transmon

To implement fast gate operations with high-fidelity, quantum processor architectures feature tunable qubit frequencies. For instance, in some cases, we need to bring two qubits into resonance to exchange (SWAP Gate) energies, while also having the capability of separating them during idling periods to minimize their interactions. To do this, we need an external parameter which allows us to access one of the degrees of freedom of the system in a controllable fashion.

One common technique is to replace the single Josephson junction with a loop interrupted by two identical junctions – forming a DC superconducting quanutm interference device (dc-SQUID). Due to the interference between the two arms of the SQUID, the effective critical current of the two parallel junctions can be decreased by applying a magnetic flux threading the loop.

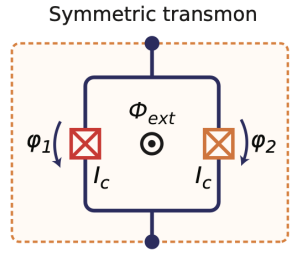

Figure 4: SQUID modular qubit circuit representation for capacitively shunted qubit modalities. Symmetric transmon qubit with Josephson energy E_J is shunted with a capacitor yielding a charging energy E_C.

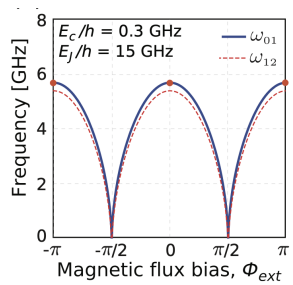

Figure 5: Corresponding qubit transition frequencies for the two lowest energy states as a function of applied magnetic flux in units of Φ_0.

The algebraic sum of branch flux of all the inductive elements along the loop plus the externally applied flux equals an integer number of superconducting flux quanta to match the fluxoid quantization condition. That is,

$$\varphi_1 - \varphi_2 + 2\varphi_e = 2\pi k$$

where $\varphi_e = \pi\Phi_{\text{ext}}/\Phi_0$. Using this condition, we can eliminate one degree of freedom and treat the SQUID-loop as a single junction, but with the important modification that E_J is tunable (via the SQUID critical current) by means of the external flux Φ_{ext}. The effective Hamiltonian of the so-called "split transmon" is then

$$H = 4E_C n^2 - 2E_J|\cos(\varphi_e)|\cos(\phi) \tag{10}$$

$$H = 4E_C n^2 - E'_J(\varphi_e)\cos(\phi) \tag{11}$$

This equation is analogous to Equation 8. with E_J replaced by $E'_J(\varphi_e) = 2E_J|\cos(\varphi_e)|$. The magnitude of the net, effective Josephson energy E'_J has a period of Φ_0 in applied flux and spans from 0 to its maximum value $2E_J$. Therefore, the qubit frequency can be tuned periodically with Φ_{ext}, shown in Figure 5.

However, the split transmon also introduces sensitivity to random flux fluctuations, known as *flux noise*. At any point, the slope of the qubit spectrum, $\partial\omega_q/\partial\Phi_{\text{ext}}$ indicates to first order how strongly this noise affects the qubit frequency. This sensitivity is only 0 at multiples of the flux quantum, $\Phi_{\text{ext}} = k\Phi_0$ where k is an integer, $\partial\omega_q/\partial\Phi_{\text{ext}} = 0$.

One method of reducing flux noise sensitivity while maintaining sufficient tunability is to make the two junctions in the split transmon *asymmetric*, which yields the following Hamiltonian

$$H = 4E_C n^2 - E_{J\Sigma}\sqrt{\cos^2(\varphi_e) + d^2\sin^2(\varphi_e)}\cos(\phi) \tag{12}$$

where $E_{J\Sigma} = E_{J1} + E_{J2}$ and $d = (\gamma - 1)/(\gamma + 1)$ is the junction asymmetry parameter, with $\gamma = E_{J2}/E_{J1}$. Going from symmetric to asymmetric transmons does not change the circuit topology, but has a profound impact for practical applications. As you can see below, the flux sensitivity is

suppressed across the entire tunable frequency range for asymmetric circuits relative to symmetric ones.

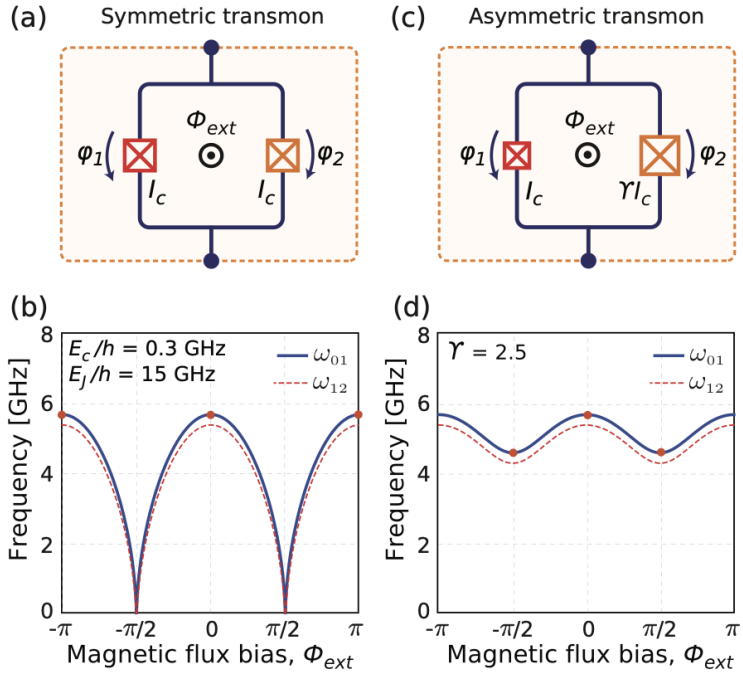

Figure 6: Symmetric transmon qubit from Figure 5 compared to an Asymmetric transmon qubit with junction asymmetry $\gamma = E_{J2}/E_{J1} = 2.5$.

1.2.2 Flux Qubit and Fluxonium

Split transmon qubits, whether symmetric or antisymmetric, still have the same topology yielding a sinusoidal potential. The degree to which properties of these qubits can be engineered has not changed. The limited anharmonicity in transmon-type qubits intrinsically causes significant residual excitation to higher-energy states, undermining fidelity of gate operations. It's necessary to introduce more complexity in the circuit.

The *flux qubit* is engineered by making the qubit loop interrupted by three or four junctions.

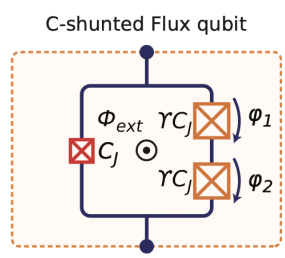

Figure 7: Capacitively shunted flux qubit, where a small principle junction (red) is shunted with a capacitor yielding a charging energy E_C.

On one branch is a smaller junction, and on the other branch are two identical junctions, both a factor γ larger in size. The addition of one more junction is non-trivial as it changes the circuit topology and reshapes the potential energy profile.

Each junction is associated with a *phase variable* and the fluxoid quantization condition again allows us to eliminate one degree of freedom. Now, we have a two-dimensional potential landscape, which compared to the simpler topology of the transmon, complicates the problem both conceptually and computationally. Under a *quasi-1D approximation*, the Hamiltonian is

$$H \approx 4E_C n^2 - E_J \cos(2\pi + \varphi_e) - 2\gamma E_J \cos(\varphi_e) \tag{13}$$

Note the phase variable in Equation 13. is the sum of the branch phases across the two array junctions, $\phi = (\varphi_1 + \varphi_2)/2$, assuming same current direction across φ_1 and φ_2. The final potential

profile as well as the corresponding energy eigenstates depends on both the external flux φ_e and the junction area ratio γ.

Most commonly, the optimal working point is when $\varphi_e = \pi + 2\pi k$, where k is an integer – that is when half a superconducting flux quantum threads the loop. Here, the qubit spectrum reaches its minimum, referred to as the *flux degeneracy point*, where flux qubits tend to have the optimal coherence time. Here, the potential energy may assume a single-well ($\gamma \geq 2$) or a double well ($\gamma < 2$) profile.

Compared to the flux qubit, which usually contains two or three array junctions, the number of array junctions in the *fluxonium* qubit dramatically increases. With the same quasi-1D approximation, the last term in Equation 13 becomes $-N_\gamma E_J \cos(\phi/N)$, giving

$$H \approx 4E_C n^2 - E_J \cos(2\phi + \varphi_e) - N\gamma E_J \cos(\phi/N) \tag{14}$$

where N denotes the number of array junctions. For large N, the argument in the cosine term, ϕ/N becomes sufficiently small that a second order expansion is a good approximation. This results in the final fluxonium Hamiltonian,

$$H \approx 4E_C n^2 - E_J \cos(\phi + \varphi_e) + \frac{1}{2} E_L \phi^2 \tag{15}$$

where $E_L = (\gamma/N)E_J$ is the inductive energy of the *effective* inductance contributed by the junction array – often known as superinductance. The potential energy is thus a quadratic term modulated by a sinusoidal term, similar to that of an rf-SQUID type flux qubit. However, the kinetic inductance of the Josephson Junction array is in general much larger than the geometric inductance of the wire in an rf-SQUID.

Depending on the relative magnitude of E_J and E_L, the fluxonium system could involve plasmon states (in the same well) and fluxon states (in different wells).